

TD — Backpropagation on a Minimal Network

Setup

We consider a two-layer linear network:

$$x \in \mathbb{R}^2 \xrightarrow{W_1, b_1} h \in \mathbb{R}^2 \xrightarrow{W_2, b_2} y \in \mathbb{R}^2$$

Parameters:

$$W_1 \in \mathbb{R}^{2 \times 2}, \quad b_1 \in \mathbb{R}^2, \quad W_2 \in \mathbb{R}^{2 \times 2}, \quad b_2 \in \mathbb{R}^2$$

Loss (sum of outputs):

$$L = y_1 + y_2 = \mathbf{1}^\top y, \quad \mathbf{1} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

1. Forward Pass

Layer 1 — hidden h

$$h = W_1 x + b_1$$
$$\begin{pmatrix} h_1 \\ h_2 \end{pmatrix} = \begin{pmatrix} W_{11}^{(1)} & W_{12}^{(1)} \\ W_{21}^{(1)} & W_{22}^{(1)} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} b_1^{(1)} \\ b_2^{(1)} \end{pmatrix}$$

Layer 2 — output y

$$y = W_2 h + b_2$$
$$\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} W_{11}^{(2)} & W_{12}^{(2)} \\ W_{21}^{(2)} & W_{22}^{(2)} \end{pmatrix} \begin{pmatrix} h_1 \\ h_2 \end{pmatrix} + \begin{pmatrix} b_1^{(2)} \\ b_2^{(2)} \end{pmatrix}$$

Loss

$$L = y_1 + y_2$$

2. Backward Pass

We apply the chain rule backwards: $y \rightarrow h \rightarrow x$, computing the gradient with respect to each parameter along the way.

Step 1 — Gradient w.r.t. y

$L = y_1 + y_2$ is linear in y , so:

$$\frac{\partial L}{\partial y} = \begin{pmatrix} \frac{\partial L}{\partial y_1} \\ \frac{\partial L}{\partial y_2} \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Step 2 — Gradient w.r.t. b_2

Since $y = W_2 h + b_2$, a unit change in b_2 propagates directly to y ($\partial y / \partial b_2 = I$), so:

$$\frac{\partial L}{\partial b_2} = \frac{\partial L}{\partial y} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Step 3 — Gradient w.r.t. W_2

For $y = W_2 h + b_2$, the general result (derived from the chain rule element-by-element) gives:

$$\frac{\partial L}{\partial W_2} = \frac{\partial L}{\partial y} \cdot h^\top$$

$$\frac{\partial L}{\partial W_2} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} (h_1 \ h_2) = \begin{pmatrix} h_1 & h_2 \\ h_1 & h_2 \end{pmatrix} \in \mathbb{R}^{2 \times 2}$$

Dimension check: $\underbrace{\frac{\partial L}{\partial y}}_{2 \times 1} \cdot \underbrace{h^\top}_{1 \times 2} = \underbrace{\frac{\partial L}{\partial W_2}}_{2 \times 2} \checkmark$

Step 4 — Gradient w.r.t. h (backprop through layer 2)

Using the general result $\partial L / \partial x = W^\top (\partial L / \partial y)$ for a layer $y = Wx + b$:

$$\frac{\partial L}{\partial h} = W_2^\top \frac{\partial L}{\partial y} = W_2^\top \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} W_{11}^{(2)} + W_{21}^{(2)} \\ W_{12}^{(2)} + W_{22}^{(2)} \end{pmatrix} \in \mathbb{R}^2$$

Dimension check: $\underbrace{W_2^\top}_{2 \times 2} \cdot \underbrace{\frac{\partial L}{\partial y}}_{2 \times 1} = \underbrace{\frac{\partial L}{\partial h}}_{2 \times 1} \checkmark$

Step 5 — Gradient w.r.t. b_1

Since $h = W_1 x + b_1$, by the same argument as b_2 :

$$\frac{\partial L}{\partial b_1} = \frac{\partial L}{\partial h} = W_2^\top \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Step 6 — Gradient w.r.t. W_1

$$\frac{\partial L}{\partial W_1} = \frac{\partial L}{\partial h} \cdot x^\top = W_2^\top \begin{pmatrix} 1 \\ 1 \end{pmatrix} x^\top \in \mathbb{R}^{2 \times 2}$$

Expanding explicitly:

$$\frac{\partial L}{\partial W_1} = \begin{pmatrix} W_{11}^{(2)} + W_{21}^{(2)} \\ W_{12}^{(2)} + W_{22}^{(2)} \end{pmatrix} \begin{pmatrix} x_1 & x_2 \end{pmatrix} = \begin{pmatrix} (W_{11}^{(2)} + W_{21}^{(2)})x_1 & (W_{11}^{(2)} + W_{21}^{(2)})x_2 \\ (W_{12}^{(2)} + W_{22}^{(2)})x_1 & (W_{12}^{(2)} + W_{22}^{(2)})x_2 \end{pmatrix}$$

Dimension check: $\underbrace{\frac{\partial L}{\partial h}}_{2 \times 1} \cdot \underbrace{x^\top}_{1 \times 2} = \underbrace{\frac{\partial L}{\partial W_1}}_{2 \times 2} \checkmark$

Summary

Order	Parameter	Gradient
1	$\frac{\partial L}{\partial b_2}$	$\begin{pmatrix} 1 \\ 1 \end{pmatrix}$
2	$\frac{\partial L}{\partial W_2}$	$\begin{pmatrix} 1 \\ 1 \end{pmatrix} h^\top$
3	$\frac{\partial L}{\partial b_1}$	$W_2^\top \begin{pmatrix} 1 \\ 1 \end{pmatrix}$
4	$\frac{\partial L}{\partial W_1}$	$W_2^\top \begin{pmatrix} 1 \\ 1 \end{pmatrix} x^\top$

Key pattern: the gradient w.r.t. a weight matrix W_k is always an *outer product* of the upstream gradient (flowing into the layer's output) with the layer's input. The gradient flows backwards through $W^\top$ at each step.